Yunfeng Zhang

CURRICULUM VITAE

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Research Interests

Areas: Harmonic analysis, representation theory, analytic number theory, and dispersive partial differential equations.

Topics: Bounds for spherical functions and joint eigenfunctions of invariant differential operators on symmetric spaces; bounds for Laplace eigenfunctions on manifolds; bounds for Hecke eigenforms; Strichartz estimates for the Schrödinger equation; and well-posedness and long-time behavior of nonlinear dispersive equations.

Education

Ph.D. in Mathematics, University of California, Los Angeles 2012–2018
Advisors: Rowan Killip and Monica Visan

B.S. in Mathematics, Tsinghua University 2008–2012

Academic Appointments

University of Mississippi 2026–present
Visiting Assistant Professor

University of Cincinnati 2024–2026
Visiting Assistant Professor

Peking University 2021–2024
TAL Assistant Professor

University of Connecticut 2018–2021
Assistant Research Professor

Honors and Awards

UCLA Mathematics Graduate Research Presentation Prize 2018

Tsinghua University Outstanding Graduate Award 2012

Fellowship in the Tsinghua Xuetang Mathematics Program 2009–2012

Grants

Co-I (30% effort), National Key Research and Development Program of China 2022–2027
Geometry and Analysis of Homogeneous Spaces
Total award: CNY 3,000,000

PI, Fundamental Research Funds for the Central Universities, Peking University 2021–2023
Analysis on Lie Groups
Total award: CNY 200,000

Mentoring

An Nguyen, undergraduate student at the University of Cincinnati 2025–2026

Research Publications

10. **Pointwise character bounds for $\mathrm{SU}(3)$**
Bulletin of the London Mathematical Society 58 (2026), No. 10, Article ID e70519. arXiv:2604.17589

9. **Growth of Fourier–Lebesgue norms for mKdV**
With Saikatul Haque, Rowan Killip and Monica Visan.
Partial Differential Equations and Applications 7 (2026), No. 4, Article No. 31. arXiv:2511.17471

8. **Restriction of eigenfunctions on products of spheres to submanifolds of maximal flats**
Proceedings of the American Mathematical Society 154 (2026), No. 9, 3921–3931. arXiv:2511.14615

7. **Local well-posedness for nonlinear Schrödinger equations on compact product manifolds**
Journal of Differential Equations 467 (2026), Article ID 114288. arXiv:2503.09442

6. **Bounds of restriction of characters to submanifolds**
Mathematische Zeitschrift 312 (2026), No. 1, Article No. 13. arXiv:2402.03178

5. **Global well-posedness and equicontinuity for modified Korteweg–de Vries equations in modulation spaces**
With Saikatul Haque, Rowan Killip and Monica Visan.
Pure and Applied Analysis 7 (2025), No. 3, 615–637. arXiv:2411.05300

4. **On Fourier restriction type problems on compact Lie groups**
Indiana University Mathematics Journal 72 (2023), No. 6, 2631–2699. arXiv:2005.11451

3. **Schrödinger equations on compact globally symmetric spaces**
Journal of Geometric Analysis 31 (2021), No. 11, 10778–10819. arXiv:2005.00429

2. **Strichartz estimates for the Schrödinger equation on products of odd-dimensional spheres**
Nonlinear Analysis 199 (2020), Article ID 112052. arXiv:2301.02823

1. **Strichartz estimates for the Schrödinger flow on compact Lie groups**
Analysis & PDE 13 (2020), No. 4, 1173–1219. arXiv:1703.07548

Research Preprints

3. **Rootwise estimates for Weyl alternants**
With Xiaocheng Li. arXiv:2609.32214

2. **Sharp bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy critical NLS**
With Yangkendi Deng and Zehua Zhao. arXiv:2509.09565

1. **Algebraic and analytic properties of invariant differential operators on a homogeneous space of complexity one**
With Hanlong Fang and Xiaocheng Li. arXiv:2301.00529

Expository Publications

1. **Analysis on compact symmetric spaces: eigenfunctions and nonlinear Schrödinger equations**
In: *Trends in Mathematics, Research Perspectives Ghent Analysis and PDE Center 3* (2024), 235–240.

Talks

* *denotes a contributed talk; unmarked talks are invited.*

2026

Oct. **Rootwise estimates for Weyl alternants**[*]
Palmetto Number Theory Series (PANTS) XLI, Georgia Institute of Technology

Apr. **Bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy-critical NLS**
Applied Analysis Seminar, Louisiana State University

Apr. **Pointwise character bounds: the case of** $\mathrm{SU}(3)$
Analysis Seminar, University of Cincinnati

Mar. **Bilinear eigenfunction estimate, anisotropic Strichartz estimate, and energy-critical NLS**[*]
The Fifteenth Ohio River Analysis Meeting, University of Kentucky

2025

Nov. **Well-posedness of the energy-critical NLS on** $\mathbb{R} \times \mathbb{S}^3$
Beijing Institute of Technology

Oct. **Well-posedness of the energy-critical NLS on** $\mathbb{R} \times \mathbb{S}^3$
Analysis Seminar, Bielefeld University

Jun. **On NLS posed on** $\mathbb{R} \times \mathbb{S}^3$
Workshop on Dispersive PDEs and Control Theory, Beijing Institute of Technology

May **Bounds of restriction of characters to submanifolds**
Tsinghua University

Apr. **The modified KdV equation beyond Sobolev spaces**
Analysis Seminar, University of Cincinnati

Mar. **Bounds of restriction of characters to submanifolds**
AMS Special Session on Recent Trends in Harmonic Analysis and PDE, University of Kansas

Mar. **Multi-linear multi-parameter eigenfunction bounds and NLS on compact manifolds**
Beijing Institute of Technology

2024

Dec. **On the modified KdV equation in modulation spaces**
Joint Meeting of the NZMS, AustMS and AMS: Special Session on Harmonic Analysis and Hamiltonian PDE, University of Auckland

Oct. **Semiclassical fun with** $\mathrm{SU}(3)$[*]
Prairie Analysis Seminar 2024, University of Kansas

Sep. **Semiclassical fun with** $\mathrm{SU}(3)$
Analysis Seminar, University of Cincinnati

Jun. **Bounds of restriction of characters to submanifolds**
Analysis Seminar, Southern University of Science and Technology

Jun. **The modified KdV in modulation spaces: conservation laws and equicontinuity of solutions**
Beijing Institute of Technology

May **Bounds of restriction of characters to submanifolds**
Analysis Seminar, University of Wisconsin–Madison

Jan. **Bounds of restriction of characters to submanifolds**
Beijing Institute of Technology

2023

Dec. **Harmonic analysis on compact symmetric spaces**[*]
Global Young Scholars Forum, Beijing Normal University

Dec. L^p **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Young Scholars Forum, ShanghaiTech University

Dec. L^p **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Young Mathematician Forum, Shanghai Jiao Tong University

Dec. **Harmonic analysis on compact symmetric spaces**[*]
Vision Forum for International Young Scholars, Beihang University

Nov. L^p **norms of Laplacian eigenfunctions on compact symmetric spaces**[*]
Global Forum for Young Mathematicians, Southern University of Science and Technology

Nov. L^p **norms of Laplacian eigenfunctions on compact Lie groups**[*]
Teli Forum for International Young Scholars, Beijing Institute of Technology

Sep. **Discrete Fourier restriction and the Kloosterman circle method**
Colloquium, Huaibei Normal University

Sep. **Fourier restriction type problems on compact Lie groups**
Beijing Institute of Technology

2021

Nov. **Nonlinear Schrödinger equation on compact symmetric spaces**
Methusalem Junior Analysis & PDE Seminar, Ghent University

Oct. **Fourier restriction bounds on compact symmetric spaces**
Conference on Harmonic Analysis and Symmetric Spaces, University of Wisconsin–Madison

Apr. **Strichartz estimate for the Schrödinger equation on compact globally symmetric spaces**
Oberseminar Analysis, Bielefeld University

Feb. **Schrödinger equations on compact globally symmetric spaces**
Weekly Seminar on Geometric and Functional Inequalities and Applications, University of Connecticut

2020

Sep. **Size of Laplacian eigenfunctions on compact symmetric spaces**
AMS Special Session on Geometric Inequalities and Nonlinear PDEs, University of Texas at El Paso

Mar. **Strichartz estimates for the Schrödinger equation on compact symmetric spaces**
AMS Special Session on Analysis on Homogeneous Spaces, Tufts University (Cancelled due to COVID)

Service and Outreach

Co-organizer of the minisymposium "Recent Developments in Applied Harmonic Analysis" at the 11th Annual Meeting of the SIAM Central States Section, Jackson State University (2026)

Judge for the 40th (2025) and 41st (2026) Annual UC Math Bowl — a mathematics competition for middle and high school students hosted by the University of Cincinnati

Co-organizer of the Analysis and Probability Seminar at the University of Connecticut, Fall 2020 and Spring 2021

Reviewer for Mathematical Reviews and zbMATH Open

Referee for
Beijing Journal of Pure and Applied Mathematics
Bulletin of the London Mathematical Society
Communications on Pure and Applied Analysis
Journal of Differential Equations
Journal of Functional Analysis
Journal of Pseudo-Differential Operators and Applications
Pacific Journal of Mathematics
Selecta Mathematica (quick opinion)
Transactions of the American Mathematical Society

Teaching Experience

AS INSTRUCTOR

University of Mississippi

Elementary Differential Equations	Fall 2026
Unified Calculus and Analytic Geometry III — honors section	Fall 2026
Unified Calculus and Analytic Geometry III — standard section	Fall 2026

University of Cincinnati

Linear Algebra	Spring 2026
Calculus II (two sections)	Fall 2025
Pre Calculus	Fall 2025
Calculus I	Spring 2025
Applied Calculus I	Spring 2025
College Algebra (two sections)	Fall 2024

Peking University

Linear Algebra B ("B" stands for "for the Physical Sciences")	Fall 2023
Linear Algebra B	Fall 2022
Advanced Mathematics B (i.e. Calculus for the Physical Sciences)	Fall 2021

University of Connecticut

Partial Differential Equations (two classes)	Spring 2021
Partial Differential Equations (two classes)	Fall 2020
Axiomatic Geometry (two classes)	Spring 2020
Introduction to Complex Variables (two classes)	Fall 2019
Partial Differential Equations (two classes)	Spring 2019
Honors Calculus II	Fall 2018
Honors Multivariable Calculus	Fall 2018

University of California, Los Angeles

Calculus for Life Sciences Students II	Summer 2017

AS TEACHING ASSISTANT

University of California, Los Angeles

Probability Theory II	Spring 2018, Spring 2017, Winter 2017, Winter 2016
Algebra for Applications	Winter 2018
Analysis I	Fall 2017, Winter 2016, Fall 2015
Probability Theory I	Winter 2017, Winter 2015
Differential and Integral Calculus	Fall 2016
Linear & Nonlinear Systems of Differential Equations	Fall 2015, Spring 2015, Winter 2014
Mathematical Game Theory	Summer 2015
Partial Differential Equations	Spring 2015
Discrete Structures	Winter 2015
Precalculus	Fall 2014, Fall 2012
Calculus for Life Sciences Students I	Fall 2014
Linear Algebra I	Summer 2014
Differential Geometry II	Spring 2014
Ordinary Differential Equations	Spring 2014, Winter 2014
Integration and Infinite Series	Fall 2013
Complex Analysis for Applications	Spring 2013
Differential Equations	Winter 2013